Quantum Information Processing

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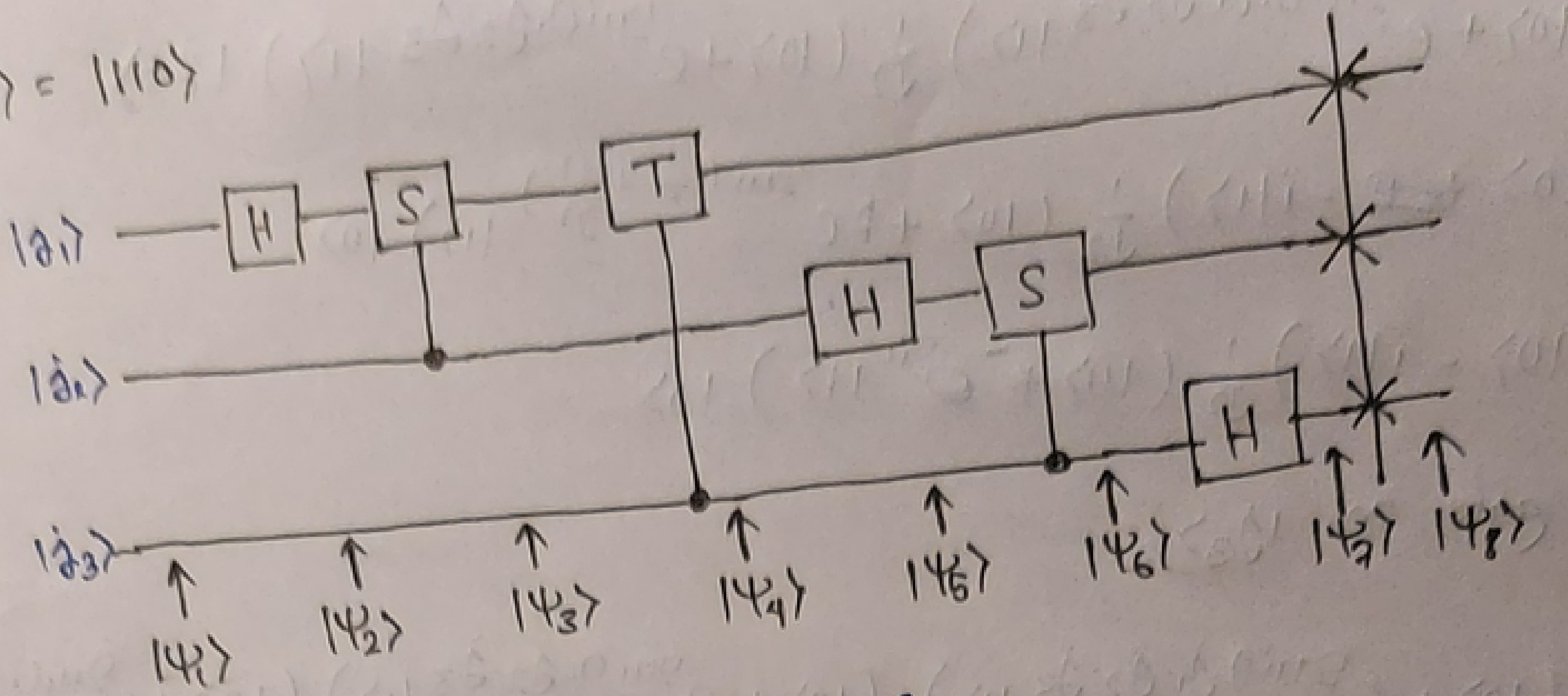
Assignment-3

1. $|\psi_1\rangle = |110\rangle$



In our case, $j_1 = 1$; $j_2 = 1$; $j_3 = 0$

$|\psi_1\rangle = |j_1 j_2 j_3\rangle = |j_1\rangle|j_2\rangle|j_3\rangle = |1\rangle|1\rangle|0\rangle$

$H|j_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.j_1}|1\rangle)$

$|\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.j_1}|1\rangle)|j_2\rangle|j_3\rangle$

Putting $j_1 = 1$, $j_2 = 1$, $j_3 = 0$

$|\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i(\frac{1}{2})}|1\rangle)|j_2\rangle|j_3\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|j_2\rangle|j_3\rangle$
$= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|1\rangle|0\rangle$

Then,

$S(H|j_1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.j_1 j_2}|1\rangle)$

So, $|\psi_3\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.j_1 j_2}|1\rangle)|j_2\rangle|j_3\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i(\frac{1}{2}+\frac{1}{2^2})}|1\rangle)|j_2\rangle|j_3\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i \frac{3}{4}}|1\rangle)|j_2\rangle|j_3\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)|j_2\rangle|j_3\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)|1\rangle|0\rangle$

$\frac{1}{2}+\frac{1}{4}$

$\frac{3}{4}$

$(e^{\pi i/2})^3$

$i^3 = -i$

Next,

$|\psi_4\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.j_1 j_2 j_3}|1\rangle)|j_2\rangle|j_3\rangle$ after using the T gate

$= \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i(\frac{1}{2}+\frac{1}{2^2}+\frac{0}{2^3})}|1\rangle)|j_2\rangle|j_3\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i \times \frac{3}{4}}|1\rangle)|j_2\rangle|j_3\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)|j_2\rangle|j_3\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)|1\rangle|0\rangle$

Next, apply H on $|j_2\rangle$

So, $|\psi_5\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.j_1 j_2 j_3}|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.j_2}|1\rangle)|j_3\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i(\frac{1}{2})}|1\rangle)|j_3\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|0\rangle$

After applying S gate on $H|j_2\rangle$ we have,

$$|\psi_6\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.j_1 j_2 j_3}|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.j_2 j_3}|1\rangle)|j_3\rangle$$

$$= \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i(\frac{1}{2}+\frac{0}{2^2})}|1\rangle)|0\rangle$$

$$= \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|0\rangle$$

Now Apply H gate on $|j_3\rangle$,

$$|\psi_7\rangle = \frac{1}{2^{3/2}}(|0\rangle + e^{2\pi i 0.j_1 j_2 j_3}|1\rangle)(|0\rangle + e^{2\pi i 0.j_2 j_3}|1\rangle)(|0\rangle + e^{2\pi i 0.j_3}|1\rangle)$$

$$= \frac{1}{2^{3/2}}(|0\rangle - i|1\rangle)(|0\rangle - |1\rangle)(|0\rangle + |1\rangle)$$

After SWAP.

$$|\psi_8\rangle = \frac{1}{2^{3/2}}(|0\rangle + e^{2\pi i 0.j_3}|1\rangle)(|0\rangle + e^{2\pi i 0.j_2 j_3}|1\rangle)(|0\rangle + e^{2\pi i 0.j_1 j_2 j_3}|1\rangle)$$

$$\cancel{= \frac{1}{2^{3/2}}(|0\rangle + e^{2\pi i 0.\cdots}}$$

$$= \frac{1}{2^{3/2}}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)(|0\rangle - i|1\rangle)$$

$$= \frac{1}{2^{3/2}}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)(|0\rangle - i|1\rangle)$$

$$= \frac{1}{2^{3/2}}(|000\rangle - i|001\rangle - |010\rangle + i|011\rangle + |100\rangle - i|101\rangle - |110\rangle + i|111\rangle)$$

Final Ans.